HCI

Human and Computer Interaction

WHY HCI??????????????



Why HCI is offered in 7th semester of AI

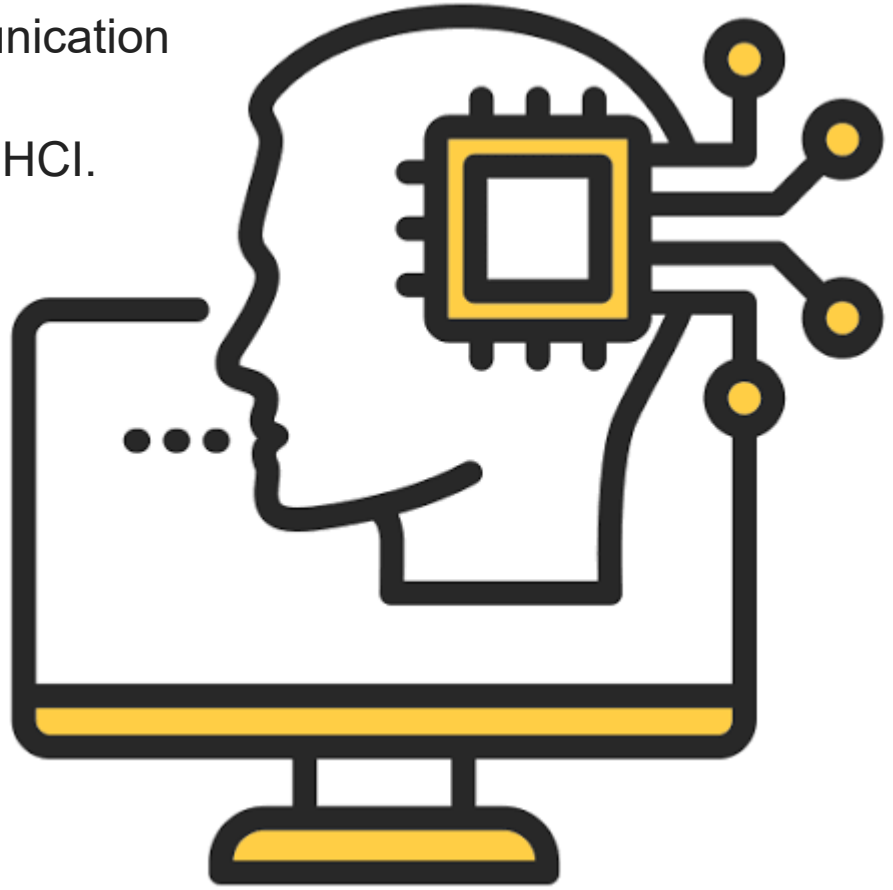
The key message for students should be:

“You already know how to build a system. HCI teaches you how to build a system that people can actually use.”

HCI principles Human-Computer Interaction (HCI)

Human + Computer = Communication

Good communication = Good HCI.



For an AI student, this is particularly important because an AI model can be technically accurate but still be a **bad product**.

For example:
AI model: Detects diseases from X-rays with 95% accuracy.
HCI problem: Doctor cannot understand the result, doesn't know why the AI made its prediction, or the interface takes too long to use.
So HCI connects **AI + humans + interface + usability**.

How HCI helps in FYP

You can repeatedly connect every topic to FYP:

FYP idea: AI-based depression-support chatbot

HCI helps students ask:

- Who is going to use it?
- What does the user need?
- Is the conversation understandable?
- Is the interface confusing?
- Where should the chatbot display its response?
- What happens if the AI gives an incorrect answer?
- How do we make the interface accessible?
- How do we test whether users can actually use it?

Another example:

AI-based face-recognition attendance system
Without HCI → *“Our model has 97% accuracy.”*
With HCI → *“A teacher can mark attendance in three clicks, immediately identify recognition errors, manually correct them, and understand why a student was not recognized.”*

Suggested HCI course outline

1. Introduction to HCI
2. Why HCI matters in AI and software development
3. Human factors and users
4. Interaction and interface basics
5. User-centered design
6. Usability and user experience (UX) /UI
7. Human perception, memory, attention and cognition
8. Interaction styles and design principles
9. GUI and mobile interfaces
10. Information architecture and navigation
11. Prototyping and wireframing
12. Usability testing and evaluation
13. Accessibility and inclusive design
14. HCI for AI systems
15. Conversational interfaces and chatbots
16. HCI in FYP/project development

HCI Course Outline (16 Weeks)

- 1 Introduction to HCI, Why HCI Matters, Course Overview
- 2 History and Evolution of HCI
- 3 Human Factors (Memory, Attention, Perception, Psychology)
- 4 Interaction Styles (CLI, GUI, Touch, Voice, Gesture, VR/AR)
- 5 User-Centered Design (UCD)
- 6 Design Principles (Visibility, Feedback, Consistency, Affordance)
- 7 Color Theory, Typography & Layout
- 8 Midterm + Prototype Review
- 9 Usability Principles & Nielsen's Heuristics
- 10 User Research (Interviews, Personas, Scenarios)

HCI Course Outline (16 Weeks)

- 11 Wireframing and Prototyping (Figma)
- 12 Accessibility and Inclusive Design
- 13 Mobile App & Web Interface Design
- 14 AI and HCI (ChatGPT, Voice Assistants, Recommendation Systems)
- 15 Emerging Technologies (AR, VR, Brain-Computer Interfaces)
- 16 Final Project Demonstration & Viva

"Which app do you use immediately after waking up?"

- WhatsApp
-
- Instagram
-
- TikTok
-
- ChatGPT
-
- Google Maps
-
- YouTube

Why do you prefer this app?

- Easy to use
-
- Beautiful
-
- Fast
-
- Attractive

Why is Google Search easier than Yahoo?

- Why is Netflix addictive?
-
- Why is ChatGPT simple to use?
-
- Why does Instagram keep us scrolling?

Answer ?

- The secret isn't only programming.
- It is Human-Computer Interaction.

Real-Life Examples

- ATM Machine
- Foodpanda
- Uber
- WhatsApp
- YouTube

"What makes these apps easy?"



Your Answer?

- Big buttons
- Clear icons
- Fast navigation
- Simple colors
- Easy menus

Why AI Students Must Learn HCI

- But...
-
- Nobody can use it.
-
- Buttons are confusing.
-
- Voice doesn't work.
-
- Navigation is terrible.
-
- Then your AI fails.

For Example

- ChatGPT
-
- Tesla Dashboard
-
- Apple Siri
-
- Alexa
-
- Google Assistant

Fun Activity

- One app You love.
- One app You hate.
- Why?
- Connect HCI with AI
- Write your answers on the board.

Textbook: A good primary textbook is **Human-Computer Interaction: An Empirical Research Perspective by I. Scott MacKenzie**. For an easier, student-friendly reference, **Interaction Design: Beyond Human-Computer Interaction by Helen Sharp, Yvonne Rogers, and Jenny Preece** is excellent.

Still Confused?

